Artificial Intelligence

First-Order Logic

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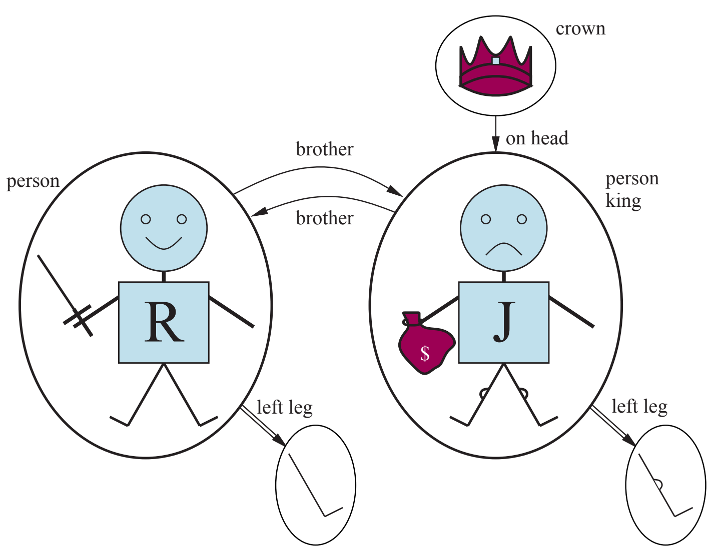
1 Representation

Language	Ontological Commitment (What exists in the world)	Episteme (What ar
Propositional logic	facts	true/false
First-order logic	facts, objects, relations	true/false
Temporal logic	facts, objects, relations, times	true/false
Probability theory	facts	degree o
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known in

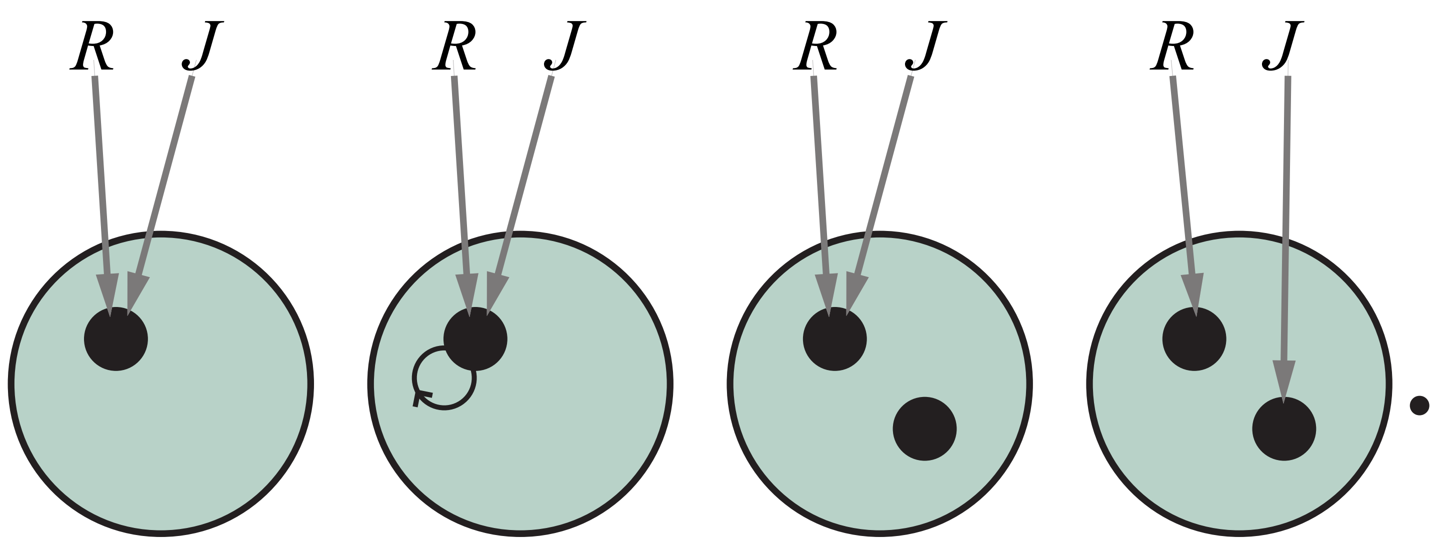
2 Syntax and Semantics of First-Order Logic



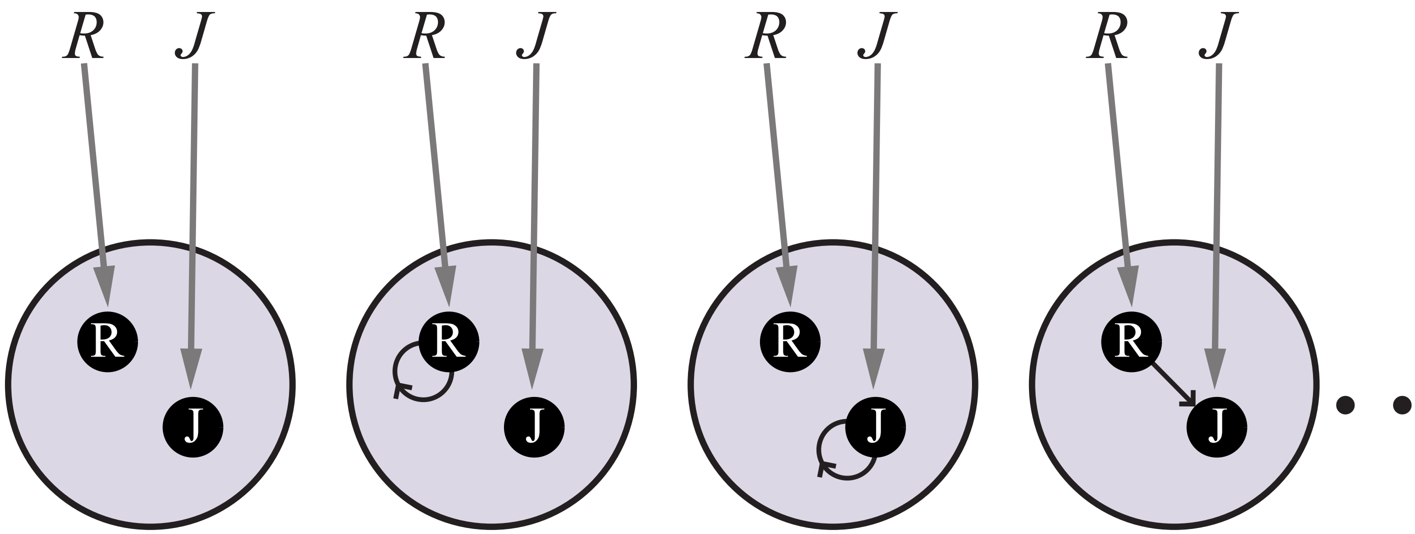
3 Syntax and Semantics of First-Order Logic

$$
\begin{array}{rcl}
\textit{Sentence} & \rightarrow & \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} & \rightarrow & \textit{Predicate} \mid \textit{Predicate}(\textit{Term},\ldots) \mid \textit{Term} = \textit{Term} \\
\textit{ComplexSentence} & \rightarrow & (\ \textit{Sentence}\ ) \\
 & \mid & \neg\ \textit{Sentence} \\
 & \mid & \textit{Sentence} \wedge \textit{Sentence} \\
 & \mid & \textit{Sentence} \vee \textit{Sentence} \\
 & \mid & \textit{Sentence}\ \Rightarrow\ \textit{Sentence} \\
 & \mid & \textit{Sentence}\ \Leftrightarrow\ \textit{Sentence} \\
 & \mid & \textit{Quantifier}\ \textit{Variable},\ldots\ \textit{Sentence} \\
 & & \\
\textit{Term} & \rightarrow & \textit{Function}(\textit{Term},\ldots) \\
 & \mid & \textit{Constant} \\
 & \mid & \textit{Variable} \\
 & & \\
\textit{Quantifier} & \rightarrow & \forall \mid \exists \\
\textit{Constant} & \rightarrow & A \mid X_1 \mid \textit{John} \mid \cdots \\
\textit{Variable} & \rightarrow & a \mid x \mid s \mid \cdots \\
\textit{Predicate} & \rightarrow & \textit{True} \mid \textit{False} \mid \textit{After} \mid \textit{Loves} \mid \textit{Raining} \mid \cdots \\
\textit{Function} & \rightarrow & \textit{Mother} \mid \textit{LeftLeg} \mid \cdots \\
\text{OPERATOR PRECEDENCE} & : & \neg, =, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{array}
$$

4 Syntax and Semantics of First-Order Logic



5 Syntax and Semantics of First-Order Logic



6 Using First-Order Logic

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7 Knowledge Engineering in First-Order Logic